EVERY RECURSIVELY ENUMERABLE LANGUAGE-SOUND EXTENSION OF IDENTITY-FREE FIXED-POINT SEMIRING THEORY IS SIGMA01-COMPLETE

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Editorial note. Source https://arxiv.org/abs/2512.19401v1. Copyright Anupam Das, Abhishek De and Stepan L. Kuznetsov, CC BY4.0. The strongest identity-free fixed-point-semiring undecidability theorem is counted once with all complete author proof dependencies. Conference DOI10.4230/LIPIcs.FSCD.2026.15 and title Undecidability for Semirings with Fixed Points identify the same work. Only two empty-labelled inference-bar wrappers are mechanically typeset using standard fraction bars; original mathematical expressions and author source are retained.

Allowed standard prerequisites (editorial).

- Recursively enumerable sets, effective inseparability, Turing-machine configurations and context-free grammars. Native sections2/3 and cited standard e.i. separator theorem. Actual protocol grammar and new semiring algebraic derivations are retained rather than outsourced.

2. Some theories of semirings with fixed points

Let $\mathcal{A}$ be a finite alphabet and $\mathcal{V}$ be a countably infinite set of variables disjoint from $\mathcal{A}$. We consider **μ-expressions** e, f, etc. generated by the following grammar:

$$e, f, \ldots \quad ::= \quad 0 \quad | \quad 1 \quad | \quad a \in \mathcal{A} \quad | \quad x \in \mathcal{V} \quad | \quad e + f \quad | \quad e \cdot f \quad | \quad \mu x e$$

We usually just write ef instead of $e \cdot f$.

Structures interpreting μ-expressions, formally, must be defined appropriately taking account of variable binding and compatible with substitution. A formal exposition can be found in e.g. [ÉL05, Lei16], so here we give only examples:

Example 1 (Languages)**.** Temporarily expand the syntax of expressions to include each language $A \subseteq \mathcal{A}^*$ as a new constant symbol. We associate to each closed μ-expression e (of this expanded syntax) a language $\mathcal{L}(e) \subseteq \mathcal{A}^*$ as follows:

$$\begin{aligned} \mathcal{L}(0) &:= 0 \\ \mathcal{L}(1) &:= \{\varepsilon\} \\ \mathcal{L}(a) &:= \{a\} \\ \mathcal{L}(A) &:= A \\ \mathcal{L}(e + f) &:= \mathcal{L}(e) \cup \mathcal{L}(f) \\ \mathcal{L}(ef) &:= \{uv : u \in \mathcal{L}(e) \ \&\ v \in \mathcal{L}(f)\} \\ \mathcal{L}(\mu x e(x)) &:= \bigcap \{A : \mathcal{L}(e(A)) \subseteq A\} \end{aligned}$$

Every symbol is interpreted by $\mathcal{L}(\cdot)$ as a monotone operation, with respect to $(\mathcal{P}(\mathcal{A}^*), \subseteq)$. The *Knaster-Tarski* fixed point theorem tells us that monotone operations on complete lattices have least fixed points,[1] in particular with the least fixed point of $A \mapsto e(A)$ given by $\mathcal{L}(e)$ defined above: the intersection of all *prefixed points* of $A \mapsto e(A)$.

It is not hard to see that the image of $\mathcal{L}(\cdot)$ on μ-expressions is just the context-free languages (CFLs) [Gru71]. This structure will form a model of all theories considered herein, and will guide our main undecidability results. Let us point out already the following consequence:

Proposition 2 (See, e.g., [HU79, Sip96])**.** *Deciding $\mathcal{L}(e) \subseteq_? \mathcal{L}(f)$ is Π^0_1-complete.*

2.1. Semirings with fixed points.

For generality of our results, we shall work with a minimal theory of semirings with fixed points, before considering specific examples such as (idempotent) Park μ-semirings and Conway μ-semirings later.

Definition 3 (Basic axioms)**.** The equational theory $\mu\mathsf{S}$ of μ-expressions is given by all the following axioms:

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[1] Indeed, they further have a complete sublattice of fixed points, in particular including a greatest fixed point.

(i) $(0, 1, +, \cdot)$ form a semiring:

$$\begin{array}{ccc} e + (f + g) = (e + f) + g & e(fg) = (ef)g & \\ e + f = f + e & e1 = e & e(f + g) = ef + eg \\ e + 0 = e & 1e = e & (e + f)g = eg + fg \\ & e0 = 0 & \\ & 0e = 0 & \end{array}$$

(ii) $\mu x e(x)$ is a fixed point of $x \mapsto e(x)$:

$$\mu x e(x) = e(\mu x e(x))$$

Note that $\mu\mathsf{S}$ does *not* insist that $\mu x e(x)$ is a *least* fixed point, which would require non-equational axioms. Instead, notice that the axiomatisation above is finite and equational.

Remark 4 (On μ-congruence)**.** Note that $\mu\mathsf{S}$ does not admit the **μ-congruence** rule: $\dfrac{e(x) = f(x)}{\mu x e(x) = \mu x f(x)}$. This is often built into the definition of various classes of 'μ-semirings' in the literature [ÉL05, GHK13, Lei16], but we shall not exploit this here. This means that, even if two operators are (provably) equivalent, the fixed points named by their closure under μ may be different.

For instance we do not have in $\mu\mathsf{S}$ that $\mu x(a + x) = \mu x(x + a)$. This is consistent with the fact that we are not axiomatising $\mu x e(x)$ to be a *least* fixed point, but just *some* fixed point. So, e.g., we can tweak the model $\mathcal{L}$ to satisfy $\mu x(a + x) = a$ but $\mu x(x + a) = a + b$. Of course only the former is the *least* fixed point, but the resulting structure remains a model of $\mu\mathsf{S}$.

Construing expressions in a higher-order syntax, where μ is a higher-order constant rather than a binder, the lack of μ-congruence may be seen as saying that μ does not behave *extensionally*. In practice, in this work, this means we must be careful to not carry out reasoning *under* a μ unless justified.

Remark 5 (Comparison to μ-semirings)**.** A **μ-semiring**, introduced in [ÉL05], is defined just like $\mu\mathsf{S}$ except:

- μ-semirings must satisfy *strong* μ-congruence: $\forall x\, e(x) = f(x) \to \mu x e(x) = \mu x f(x)$; and,
- μ-semirings do not require $\mu x e(x)$ to be a fixed point, i.e. axiom (ii) is omitted.

Thus our system $\mu\mathsf{S}$ is, strictly speaking, incomparable with general μ-semirings, though it will still underlie extensions considered later. $\mu\mathsf{S}$ should rather be considered a basic theory of 'semirings with fixed points', namely the theory of those μ-semirings in which $\mu x e(x)$ always denotes a fixed point of $x \mapsto e(x)$, albeit not necessarily the *least* fixed point, and albeit not extensionally so (see previous Theorem 4).

We omit μ-congruence for two main reasons: (a) we do not need it to obtain our ultimate undecidability results, rendering them strictly more general; and (b) the axiomatisation $\mu\mathsf{S}$ remains purely equational, and so could be of independent interest.

We shall soon present μ-semirings equipped with an order, so as to axiomatise the *leastness* of fixed points denoted by μ, in particular satisfying the 'strong' congruence principle mentioned in the example above. Before that let us recall a class of models of $\mu\mathsf{S}$ that is covered by our results:

Definition 6 ([ÉL05])**.** A **Conway μ-semiring** is a μ-semiring further satisfying:

(1) $\mu x e(f(x)) = e(\mu x f(e(x)))$
(2) $\mu x \mu y e(x, y) = \mu x e(x, x)$

Note that the fixed point axiom (ii) is already a consequence of (1) above, by setting $f(x) := x$, and so Conway μ-semirings are also models of $\mu\mathsf{S}$. Conway μ-semirings were introduced in [ÉL05] as a minimal theory in which certain normal forms of context-free grammars may be formally obtained, in particular *Greibach* normal form. We shall not develop its metatheory any further here, as our results about Conway μ-semirings depend only on the fact that their equational theory is recursively enumerable, extends that of $\mu\mathsf{S}$, and is modelled by the structure of languages $\mathcal{L}$ from Theorem 1.

2.2. **Equational systems and context-free grammars.** Before considering further extensions of $\mu\mathsf{S}$, let us revisit the interpretation of μ-expressions as context-free languages. It is known that Conway μ-semirings can 'solve' context-free grammars (CFGs), but we shall recast (and improve) this result within $\mu\mathsf{S}$ here.

An **equational system** is a set $\mathcal{E}(\vec{X}) = \{X_i = f_i(\vec{X})\}_{i<n}$. By recursively applying *Bekić's Lemma* ([Bek84]) we can solve such systems in $\mu\mathsf{S}$:

Proposition 7. *Fix an equational system $\mathcal{E}(\vec{X}) = \{X_i = f_i(\vec{X})\}_{i<n}$. There are (closed) expressions $\vec{e}$ s.t. $\mu\mathsf{S} \vdash \mathcal{E}(\vec{e})$.*

Proof. By induction on n:

- When $n = 0$ the statement is vacuously true.

- Fix an equational system $\{X_i = f_i(\vec{X}, X_n)\}_{i \leq n}$, where $\vec{X} = X_0, \ldots, X_{n-1}$. Write $e'_n(\vec{X}) := \mu X_n f_n(\vec{X}, X_n)$. Now, by inductive hypothesis, let $\vec{e}$ be solutions of $\{X_i = f_i(\vec{X}, e'_n(\vec{X}))\}_{i<n}$, i.e. such that $\mu\mathsf{S} \vdash e_i = f_i(\vec{e}, e'_n(\vec{e}))$, for all $i < n$. It suffices now to show that $e_n := e'_n(\vec{e})$ is a solution for X_n, i.e. $\mu\mathsf{S} \vdash e_n = f_n(\vec{e}, e_n)$, for which we reason as follows:

$$\begin{array}{ll} e_n = e'_n(\vec{e}) & \text{by definition of } e_n \\ = \mu X_n f_n(\vec{e}, X_n) & \text{by definition of } e'_n \\ = f_n(\vec{e}, \mu X_n f_n(\vec{e}, X_n)) & \text{by fixed point axiom (ii)} \\ = f_n(\vec{e}, e_n) & \text{by definition of } e_n \quad \square \end{array}$$

Recalling Theorem 4, note that solutions to an equational system are not necessarily unique, as $\mu\mathsf{S}$ does not insist that μ computes *least* fixed points.

Of course, we can use the above result to 'solve' CFGs too.

Corollary 8. *Fix a CFG $\{X_i \Rightarrow f_{ij}(\vec{X})\}_{i<n,j}$. There are μ-expressions $\vec{e} = e_0, \ldots, e_{n-1}$ s.t. $\mu\mathsf{S} \vdash e_i = \sum_j f_{ij}(\vec{e})$ for all $i < n$.*

2.3. **Ordered semirings with fixed points.** Let us continue to recall certain classes of semirings over μ-expressions from the literature.

A semiring is **idempotent** if it satisfies:

$$\mathsf{idem} \ : \quad e + e = e$$

We shall write $e \leq f := e + f = f$, typically in the presence of idempotency. Note $\leq$, sometimes called the 'natural order', defines a partial order under $\mu\mathsf{S} + \mathsf{idem}$, and moreover one over which the semiring operations are monotone:

Proposition 9. *The following hold in all idempotent semirings, and so in particular are consequences of $\mu\mathsf{S} + \mathsf{idem}$:*

$$(1) \qquad \begin{array}{cc} i \leq i & e \leq f \implies e + i \leq f + i \\ e \leq f \ \& \ f \leq e \implies e = f & e \leq f \implies eg \leq fg \\ e \leq f \ \& \ f \leq g \implies e \leq g & e \leq f \implies ge \leq gf \end{array}$$

Moreover these properties hold in any *semiring, as long as i is idempotent, and in particular are consequences of $\mu\mathsf{S} + (i + i = i)$.*

Corollary 10. *The following hold in all idempotent semirings, and so in particular are consequences of $\mu\mathsf{S} + \mathsf{idem}$:*

(2)

Moreover these properties hold in any *semiring, as long as i is idempotent, and in particular are consequences of $\mu\mathsf{S} + (i + i = i)$.*

Remark 11 (Ordered μ-semirings)**.** An **ordered μ-semiring** is a μ-semiring equipped with a partial order satisfying the properties from Eq. (1) above, as well as an ordered analogue of μ-congruence, **μ-monotonicity**: $\dfrac{e(x) \leq f(x)}{\mu x e(x) \leq \mu x f(x)}$ (see [ÉK12, ÉL02] for further details).

Again, this does not necessarily hold in models of $\mu\mathsf{S} + \mathsf{idem}$, even ones closed under (strong) μ-congruence. For instance if $e(x)$ is $a + x$ and $f(x)$ is $a + b + x$, then $e(x) \leq f(x)$ is valid in all models of $\mu\mathsf{S} + \mathsf{idem}$, but $\mu x e(x)$ needs only be greater than a and $\mu x f(x)$ needs only be greater than $a + b$. Again tweaking the language model, it is consistent to interpret $\mu x e(x)$ as $\{a, b, c\}$ but $\mu x f(x)$ as $\{a, b\}$. Note that, since $e(x) \leq f(x)$ is valid in *all* models of $\mu\mathsf{S} + \mathsf{idem} + \mu$-congruence, it follows that μ-monotonicity is not even admissible for this theory.

In this work we shall only consider the 'natural order' already defined, $e \leq f := e + f = f$, before more generally structures without idempotency. It is worth recalling at this point:

Fact 12 (Idempotent ideal)**.** *The class of idempotent elements of a semiring always forms an ideal.*

In the presence of an order it is of particular interest to examine structures where μ calculates a *least* (pre)fixed point wrt. the order,

$$\mathsf{lfp} \ : \quad e(\mu x e(x)) \leq \mu x e(x) \qquad \& \qquad e(f) \leq f \implies \mu x e(x) \leq f$$

yielding an induction principle. Note that the axioms above are neither (in)equational, nor finitely many: there is an axiom for each choice of f above.

Definition 13. A **Park μ-semiring** is a μ-semiring satisfying lfp. A **Chomsky Algebra** is an idempotent Park μ-semiring whose order is the natural order.[2]

[2] In fact the final clause here is redundant: the *only* monotone partial order on an idempotent semiring is the natural order.

Chomsky Algebras were defined in [GHK13, Lei16] as *algebraically complete* idempotent semirings, while Park μ-semirings were defined axiomatically in [ÉL02, ÉL05]. Both presentations are equivalent, when construing μ-expressions as a naming convention for equational systems (see previous subsection), as was noticed already in [GHK13]. Note that the fixed point axiom (ii) is already a consequence of lfp (see [ÉL02, ÉL05]), and so Chomsky Algebras are models of $\mu\mathsf{S} + \mathsf{idem}$, and Park μ-semirings are models of $\mu\mathsf{S}$.

Again, we shall not recount the established metatheory of (idempotent) Park μ-semirings or Chomsky Algebras: what is important for our results are that they are recursively enumerable extensions of $\mu\mathsf{S}(+\mathsf{idem})$, that are still sound for $\mathcal{L}$.

3. Effective inseparability of two problems on CFGs

Before presenting our main results on the undecidability of various theories of (idempotent) semirings with fixed points, we shall take a detour through the recursion theory of problems on CFGs.

3.1. Two problems on CFGs.

For our arguments, we shall use *effective inseparability* of two algorithmic problems for context-free grammars (CFG). We shall consider only a special class of CFGs which we call *productive* ones. Let $\mathcal{A}$ be the set of all terminal symbols and let $\mathcal{V}$ be the set of all non-terminal ones. As usual we shall assume among the non-terminals a designated **start** symbol $S \in \mathcal{V}$.

Definition 14 (Productive CFGs)**.** A CFG is **productive** if, for every production rule $X \Rightarrow \alpha_1 \dots \alpha_n$, some α_i is a terminal, i.e. some $\alpha_i \in \mathcal{A}$.

A typical example of productive CFG is a Greibach grammar without ε-productions. In Greibach grammars, we always have $\alpha_1 \in \mathcal{A}$. Thus, by Greibach's normalisation theorem [Gre65], one shows that any context-free language without the empty word is generated by a productive CFG. On the other hand, the empty word may never be generated by such a grammar. In the sequel we shall restrict our attention to productive CFGs.

The first problem we are going to consider is the *totality* problem (i.e., the question whether a given productive CFG generates all non-empty words over $\mathcal{A}$). The second one is more specific. It could be called "the grammar is total for obvious reasons," or *regularly total*. Formally:

Definition 15 ((Regular) totality)**.** A productive CFG is **total**, if it generates (from S) all non-empty words over $\mathcal{A}$.

A productive CFG G is **regularly total**, if it includes two designated non-terminals $Y, T \in \mathcal{V}$ such that:

- For each $a \in \mathcal{A}$, G includes all productions of the form:
$$S \Rightarrow aYT$$
$$T \Rightarrow a$$
$$T \Rightarrow aT$$
- There is some $n > 0$ such that:
  (1) S generates all nonempty words of length $\leq n+1$;
  (2) Y generates all words of length n.

It is not hard to see that every regulary total productive CFG is total:

Proposition 16 (Regular totality implies totality)**.** *Every regularly total productive CFG is total.*

Proof. Let G be a regularly total productive CFG and let $\vec{a}$ be a nonempty word:

- If $|\vec{a}| \leq n+1$ then it is generated from S by definition of regular totality.
- If $|\vec{a}| > n+1$ then write $\vec{a} = a\vec{b}\vec{c}$, for some $a \in \mathcal{A}$, $|\vec{b}| = n$ and $|\vec{c}| > 0$. Then $S \Rightarrow aYT$ and Y generates $\vec{b}$, by assumption, and T generates $\vec{c}$ (since it trivially generates *every* nonempty word). □

The algorithmic complexity of regular totality is dual to that of totality: while the latter is known to be Π^0_1-complete, the former belongs to Σ^0_1 (and, as we shall see below, is also complete in this class).

In what follows, we tacitly identify productive CFGs with their Gödel numbers under a reasonable encoding. We write $\mathbf{R}$ for the set of (Gödel numbers of) productive CFGs which are regularly total and $\bar{\mathbf{T}}$ for the set of (Gödel numbers of) productive CFGs which are not total.

3.2. Effective inseperability by reduction.

We recall, following [Kuz21], the notion of effective inseparability and the results which allows it to be used for proving Σ^0_1-completeness of logical theories.

Fix some acceptable enumeration $\{f_n\}_{n\in\mathbb{N}}$ of the one-input partial recursive functions on $\mathbb{N}$, and let us write W_n for "the nth r.e. set," i.e. $\mathsf{dom}(f_n)$, the domain of f_n.

Definition 17. $A, B \subseteq \mathbb{N}$ are **effectively inseparable** if:

- $A \cap B = \varnothing$; and,
- there is a partial recursive function $h(x, y)$ s.t., whenever $W_u \supseteq A$ and $W_v \supseteq B$ with $W_u \cap W_v = \varnothing$, we have $h(u, v)\downarrow \notin W_u \cup W_v$.

We shall call such $h(x, y)$ a **productive function** for A, B.

Given $A, B \subseteq \mathbb{N}$, a set $C \subseteq \mathbb{N}$ **separates** A from B (or is a **separator**) if $C \supseteq A$ but nonetheless $C \cap B = \varnothing$. In the Definition above, informally, h uniformly falsifies any attempt to separate A from B by a decidable set. Intuitively, we can see $\lambda v\, h(u, v)$ here as a 'proof' that any such W_u is not decidable: it is a recursive function that computes counterexamples to the statement that W_u has an r.e. complement.

The following result is a corollary of a theorem by Myhill [Myh55], as shown in [Rog87, Exercise 11-14]:

Theorem 18. *If $A, B \subseteq \mathbb{N}$ are effectively inseparable r.e. sets and C is an r.e. set separating them then C is Σ^0_1-complete.*

Note in the Theorem above, and generally in the sequel, we shall not be precise about whether C separates A from B or vice-versa, WLoG by symmetry of the statement.

Effective inseparability can be propagated by a (very weak) kind of reduction. Given $A, B \subseteq \mathbb{N}$ with $A \cap B = \varnothing$ and $A', B' \subseteq \mathbb{N}$ with $A' \cap B' = \varnothing$, an **e.i.-reduction** from (A', B') to (A, B) is a total computable function f such that:

(1) $f(A) \subseteq A'$
(2) $f(B) \subseteq B'$

Such maps are indeed appropriate for reducing questions of effective inseperability:

Proposition 19. *Suppose $A, B \subseteq \mathbb{N}$ are effectively inseparable, and $A', B' \subseteq \mathbb{N}$ with $A' \cap B' = \varnothing$. If f is an e.i.-reduction from (A', B') to (A, B) then A' and B' are also effectively inseparable.*

Proof. The preimage of an r.e. set under a total computable mapping is itself r.e. Indeed we have $f^{-1}(W_n) = \mathsf{dom}(f_n \circ f)$, and we can even effectively compute from n the index of the r.e. function $f_n \circ f$, say by a total recursive function g, so that always $f^{-1}(W_n) = W_{g(n)}$.

Now, let h be a productive function for A, B and let us define a productive function h' for A', B' as follows:

$$h'(u, v) = f(h(g(u), g(v)))$$

Clearly h' is a partial recursive function. To show it is productive for A', B', suppose $W_u \supseteq A'$, $W_v \supseteq B'$ with $W_u \cap W_v = \varnothing$ and let us show that $h'(u, v) \notin W_u \cup W_v$. First we have:

$$\begin{array}{rcll}
W_{g(u)} & = & f^{-1}(W_u) & \text{by definition of } g \\
& \supseteq & f^{-1}(A') & \text{since } W_u \supseteq A' \\
& \supseteq & A & \text{since } f(A) \subseteq A' \\
W_{g(v)} & = & f^{-1}(W_v) & \text{by definition of } g \\
& \supseteq & f^{-1}(B') & \text{since } W_v \supseteq B' \\
& \supseteq & B & \text{since } f(B) \subseteq B' \\
W_{g(u)} \cap W_{g(v)} & = & f^{-1}(W_u) \cap f^{-1}(W_v) & \text{by definition of } g \\
& = & f^{-1}(W_u \cap W_v) & \text{since preimages preserve intersections} \\
& = & f^{-1}(\varnothing) & \text{since } W_u \cap W_v = \varnothing \\
& = & \varnothing &
\end{array}$$

From here we have:

$$\begin{array}{rcll}
h(g(u), g(v)) & \notin & W_{g(u)} \cup W_{g(v)} & \text{since } h \text{ is productive for } A, B \\
& \notin & f^{-1}(W_u) \cup f^{-1}(W_v) & \text{by definition of } g \\
& \notin & f^{-1}(W_u \cup W_v) & \text{since preimages preserve unions}
\end{array}$$

Hence $h'(u, v) = f(h(g(u), g(v)) \notin W_u \cup W_v$, as required. □

3.3. **Effective inseperability of R and $\bar{\mathbf{T}}$.** The main result of this section is the following:

Theorem 20. *The sets* **R** *and* $\bar{\mathbf{T}}$ *are effectively inseparable.*

This subsection is devoted to proving this result. We proceed via a reduction from a more well-known pair of effectively inseparable problems: halting and looping for Turing machines. Throughout this section we shall work with one-input deterministic Turing machines that include, besides the initial and final state, also a designated *looping* state q_c with rules that force the machine to remain at this state once it is reached: $(q_c, a) \Rightarrow (q_c, a, N)$ for any letter a of the tape alphabet.

We say that a machine $\mathtt{M}$ loops on an input x if $\mathtt{M}$ reaches q_c when running on x. Looping is a very specific case of non-halting. Let

$$\mathbf{C} = \{(\mathtt{M}, x) \mid \mathtt{M} \text{ loops on input } x\},$$
$$\mathbf{H} = \{(\mathtt{M}, x) \mid \mathtt{M} \text{ halts on input } x\}$$

Again, we construe both these sets as subsets of $\mathbb{N}$ by implicitly identifying pairs $(\mathtt{M}, x)$ with their codes. The following fact is folklore (see, e.g., [Rog87, Exercise 7-55d][Kuz21, Proposition 3]):

Proposition 21. *The sets* $\mathbf{C}$ *and* $\mathbf{H}$ *are effectively inseparable.*

Now we can prove the effective inseparability of $\mathbf{R}$ and $\bar{\mathbf{T}}$ by reduction, in the sense of Proposition 19, to the effective inseparability of $\mathbf{C}$ and $\mathbf{H}$.

Let us start by briefly recalling some ideas from the standard proof of Π^0_1-hardness for the totality problem for CFGs, as presented in Kozen's textbook [Koz97]. The *halting protocol* of a Turing machine $\mathtt{M}$ on a given input x (if it exists) is the word $\#\kappa_0\#\kappa_1\#\dots\#\kappa_n\#$, where κ_0 is the initial configuration q_0x, each κ_{i+1} is the configuration which is the immediate successor of κ_i, and κ_n is a final configuration. Let $\mathcal{A}$ be the alphabet in which configurations are written down (which includes the internal alphabet of $\mathtt{M}$, its set of states, and the additional symbol $\#$). We may readily define a (computable) reducing function, mapping each pair $(\mathtt{M}, x)$ to a CFG which generates all non-empty words except for the halting protocol (if it exists). Thus, this grammar is total if and only if $\mathtt{M}$ does not halt on x.

We shall modify this construction a bit, following [Kuz21], in order to take care of looping. All non-empty words which are not the halting protocol must belong to one of the following three classes:

(I) words of length greater than 1, beginning with $\#$, which cannot be even a prefix of the halting protocol: this includes words not beginning with $\#\kappa_0\#$; words where there is not a correct configuration between two $\#$'s; words with subwords $\#\kappa\#\kappa'\#$ where κ' is not the immediate successor of κ; and, most importantly, all words including q_c;

(II) words of length greater than 1, beginning with $\#$, and not ending with $\#\kappa_F\#$ where κ_F is a final configuration (such a word could be an incomplete protocol—a prefix of the halting protocol);

(III) words beginning with a letter different from $\#$, and the one-letter word $\#$.

Note that these classes are not necessarily disjoint: a word can belong to more than one of them.

Write (I)′ and (II)′ for the sets of words from (I) and (II), respectively, with the first symbol $\#$ removed. The standard construction shows that both (I)′ and (II)′ are context-free. Denote the grammars generating (I)′ and (II)′ by G_1 and G_2, respectively; we suppose that both grammars are productive (e.g., being Greibach grammars). Here it is important that (I)′ and (II)′ are ε-free. Moreover, G_1 and G_2 are computable from $(\mathtt{M}, x)$ [Kuz21, Section 2.1].

We assume G_1 and G_2 have disjoint sets of non-terminals, writing Y and Z for their start symbols, respectively. Now define the grammar $G_{\mathtt{M},x}$ obtained from the union of G_1 and G_2 along with the following additional productions:

$$
\begin{array}{ll}
S \Rightarrow aYT & \text{for each } a \in \mathcal{A} \\
T \Rightarrow aT & \text{for each } a \in \mathcal{A} \\
T \Rightarrow a & \text{for each } a \in \mathcal{A} \\
S \Rightarrow \#Y & \\
S \Rightarrow \#Z & \\
S \Rightarrow aT & \text{for each } a \in \mathcal{A} - \{\#\} \\
S \Rightarrow a & \text{for each } a \in \mathcal{A}
\end{array}
$$

Here S and T are fresh non-terminals, and S is the new start symbol of $G_{\mathtt{M},x}$. Notice that the first three productions are exactly as in Theorem 15 for regular totality.

Proposition 22. *$G_{\mathtt{M},x}$ generates all nonempty words over $\mathcal{A}$, except the halting protocol of $\mathtt{M}$ on x (if it exists).*

Proof. For words from (I) or (II) we start by firing $S \Rightarrow \#Y$ and $S \Rightarrow \#T$, respectively, and continue as in G_1 or G_2, respectively. Words from (III) are generated by the final two rules. Note that this analysis accounts for *all* words generated by runs beginning with the final four rules from the start symbol S.

It remains to verify that the halting protocol cannot be generated by runs beginning with the first rule $S \Rightarrow aYT$. However, any such run generates only words not beginning with $\#$ and words with a prefix from (I); by definition, no such word could be the halting protocol. □

We can now define an appropriate *reducing function*:

$$f\colon (\mathtt{M}, x) \mapsto G_{\mathtt{M},x} \tag{3}$$

Of course, this function is total and computable. Moreover it allows us to deduce effective inseparability of $\mathbf{R}$ and $\bar{\mathbf{T}}$ from that of $\mathbf{C}$ and $\mathbf{H}$:

Proposition 23. *The function f in (3) is an e.i.-reduction from $(\mathbf{R}, \bar{\mathbf{T}})$ to $(\mathbf{C}, \mathbf{H})$.*

Proof. Clearly $\mathbf{R} \cap \bar{\mathbf{T}} = \varnothing$. So we need to show the following two properties:

- if $(\mathtt{M}, x) \in \mathbf{C}$, then $f((\mathtt{M}, x)) = G_{\mathtt{M},x} \in \mathbf{R}$;
- if $(\mathtt{M}, x) \in \mathbf{H}$, then $f((\mathtt{M}, x)) = G_{\mathtt{M},x} \in \bar{\mathbf{T}}$.

The second property follows immediately from Theorem 22, since the halting protocol cannot be generated by $G_{\mathtt{M},x}$. The first property is more involved.

Let $\mathtt{M}$ loop on x and let $n+1$ be the length of an (incomplete) protocol of $\mathtt{M}$ on x which reaches a configuration where $\mathtt{M}$ is in state q_c. We shall show that $G_{\mathtt{M},x}$ is regularly total, according to the given n. That is, we will show that:

(1) S generates all nonempty words of length $\leq n+1$.
(2) Y generates all words of length n.

For (1), since $\mathtt{M}$ does not halt on x, there is no halting protocol, and S generates all nonempty words, by Theorem 22. In particular, S generates all words of length $\leq n+1$.

For (2), we need to show that any word $\vec{a}$ of length n belongs to $(\mathrm{I})'$. For this, even if $\#\vec{a}$ is a prefix of the (infinite) protocol of $\mathtt{M}$ on x, it includes, by choice of n, the letter q_c, and so $\#\vec{a} \notin (\mathrm{I})$. □

Now we can finally conclude the main result of this section:

Proof of Theorem 20. Follows immediately from Theorems 19, 21 and 23. □

4. Undecidability of (extensions of) $\mu\mathsf{S} + \mathsf{idem}$

This section is devoted to a proof of our first main result:

Theorem 24. *Any r.e. extension of $\mu\mathsf{S} + \mathsf{idem}$ that is sound for $\mathcal{L}$ is Σ^0_1-complete.*

We shall actually strengthen this result in the next section to avoid the need for idempotency (and even identity), by a sort of reduction, but we present this weaker version to illustrate the main ideas in a simpler setting.

4.1. Some intermediate lemmata.

To prove Theorem 24 we first establish a couple intermediate results reasoning about (productive) CFGs in $\mu\mathsf{S}(+\mathsf{idem})$. As we shall later consider settings without idempotency and identity, we shall be careful to avoid relying on the multiplicative identity 1, and shall also be sensitive to the use of idempotency.

Lemma 25. *Fix a CFG G and write e_X for (any) corresponding solutions in $\mu\mathsf{S}$ of each non-terminal X of G, cf. Theorem 8. If X generates a word $\vec{a}$ then $\mu\mathsf{S} + \mathsf{idem} \vdash \vec{a} \leq e_X$.*

Proof. By induction on the derivation tree of $\vec{a}$ from X in G. Working in $\mu\mathsf{S} + \mathsf{idem}$:

- If $X \Rightarrow \vec{a}$, then we have $\vec{a} + g = e_X$ for some g by Theorem 8, and so $\vec{a} \leq e_X$ by idempotency of $\vec{a}$ and Theorem 9.
- If $X \Rightarrow \vec{a}_0 X_1 \vec{a}_1 \cdots \vec{a}_{n-1} X_n \vec{a}_n$, s.t. $\vec{a} = \vec{a}_0 \vec{b}_1 \vec{a}_1 \cdots \vec{b}_n \vec{a}_n$, then each $\vec{b}_i$ is generated by X_i by smaller derivation trees. Thus by inductive hypothesis we have $\vec{b}_i \leq e_{X_i}$ for all i. Finally by monotonicity of products wrt $\leq$, we have indeed $\vec{a} \leq e_X$. □

Let us write $\top := \mu x \left(\sum_{a \in \mathcal{A}} a + \sum_{a \in \mathcal{A}} ax \right)$, which trivially computes the 'total' language of all nonempty words.

Lemma 26. *For $n > 0$, we have $\mu\mathsf{S} \vdash \top = \sum_{|\vec{a}|=1}^{n} \vec{a} + \sum_{|\vec{a}|=n} \vec{a}\top$.*

Proof. By induction on $n > 0$:

- When $n = 1$, then the statement follows by just an unfolding of the fixed point $\top$, axiom (ii).
- For the inductive step we have:

$$
\begin{aligned}
\top &= \sum_{|\vec{a}|=1}^{n} \vec{a} + \sum_{|\vec{a}|=n} \vec{a}\top && \text{by inductive hypothesis} \\
&= \sum_{|\vec{a}|=1}^{n} \vec{a} + \sum_{|\vec{a}|=n} \vec{a} \left(\sum_{a \in \mathcal{A}} a + \sum_{a \in \mathcal{A}} a\top \right) && \text{by unfolding } \top\text{, axiom (ii)} \\
&= \sum_{|\vec{a}|=1}^{n} \vec{a} + \sum_{|\vec{a}|=n} \vec{a} \sum_{a \in \mathcal{A}} a + \sum_{|\vec{a}|=n} \vec{a} \sum_{a \in \mathcal{A}} a\top && \text{by distributivity} \\
&= \sum_{|\vec{a}|=1}^{n} \vec{a} + \sum_{|\vec{a}|=n+1} \vec{a} + \sum_{|\vec{a}|=n+1} \vec{a}\top && \text{by distributivity} \\
&= \sum_{|\vec{a}|=1}^{n+1} \vec{a} + \sum_{|\vec{a}|=n+1} \vec{a}\top && \text{by rebracketing sums} \quad \square
\end{aligned}
$$

4.2. **Small proofs of regular totality.** Note that $\top = \mu x \left(\sum_{a \in \mathcal{A}} a + \sum_{a \in \mathcal{A}} ax \right)$ is a solution of the non-terminal T in any regularly total grammar. In what follows, whenever we fix solutions of a regularly total grammar, we assume the solution for T is $\top$. Again, we shall be sensitive to the use of idempotency, making clear any such instances for our strengthenings in the next section.

Proposition 27. *Fix a productive CFG $G \in \mathbf{R}$ and let e_X be solutions of each if its non-terminals X in $\mu\mathsf{S}$, cf. Theorem 8, with $e_T = \top$. Then $\mu\mathsf{S} + \mathsf{idem} \vdash \top \leq e_S$.*

Proof. Since $G \in \mathbf{R}$, let $n > 0$ be such that:

(1) S generates all nonempty words of length $\leq n+1$.
(2) Y generates all words of length n.
(3) G has a production $S \to aYT$ for each $a \in \mathcal{A}$.

Let us work in $\mu\mathsf{S} + \mathsf{idem}$. By (2) above and Theorem 25 we have $\vec{a} \leq e_Y$ whenever $|\vec{a}| = n$. By idempotency of e_Y and Theorem 10 we have $\sum_{|\vec{a}|=n} \vec{a} \leq e_Y$. Thus by monotonicity of $\cdot$ and distributivity:

$$\sum_{|\vec{a}|=n} \vec{a}\top \leq e_Y \top$$

Now by (3) and Theorem 8 we have $g + \sum_{a \in \mathcal{A}} a e_Y \top = e_S$ for some g (and since e_T is $\top$). So by idempotency of $\sum_{a \in \mathcal{A}} a e_Y \top$ we have:

$$\sum_{a \in \mathcal{A}} a e_Y \top \leq e_S$$

Putting the two displays above together we have $\sum_{a \in \mathcal{A}} a \sum_{|\vec{a}|=n} \vec{a}\top \leq e_S$ by monotonicity, and so by distributivity:

$$\sum_{|\vec{a}|=n+1} \vec{a}\top \leq e_S \tag{4}$$

Now by (1) and again by Theorem 25 we have $\vec{a} \leq e_S$ whenever $1 \leq |\vec{a}| \leq n+1$. Again by idempotency of e_S and monotonicity of $+$ we have $\sum_{|\vec{a}|=1}^{n+1} \vec{a} \leq e_S$. Putting this together with (4) above, we have by idempotency of e_S and Theorem 10:

$$\sum_{|\vec{a}|=1}^{n+1} \vec{a} + \sum_{|\vec{a}|=n+1} \vec{a}\top \leq e_S$$

From here we conclude by Theorem 26 and transitivity of $\leq$, Theorem 9. □

4.3. **Putting it together.** We are now ready to prove the main result of this section:

Proof of Theorem 24. Let us assume that all CFGs have start symbol S and write e_X^G for the solution of the nonterminal X in G obtained by Theorem 8. Let $\mathsf{U} \supseteq \mu\mathsf{S} + \mathsf{idem}$ be r.e. with $\mathcal{L} \models \mathsf{U}$. Writing $\mathsf{Tot}(\mathsf{U}) := \{G \text{ productive} : \mathsf{U} \vdash \top \leq e_S^G\}$, we have:

- $\mathsf{Tot}(\mathsf{U}) \cap \bar{\mathbf{T}} = \varnothing$, by assumption that $\mathcal{L} \models \mathsf{U}$.
- $\mathbf{R} \subseteq \mathsf{Tot}(\mathsf{U})$, since $\mu\mathsf{S} + \mathsf{idem} \subseteq \mathsf{U}$ and by Theorem 27.

Thus by Theorems 18 and 20 we have that $\mathsf{Tot}(\mathsf{U})$ is Σ_1^0-complete. Since there is an obvious computable reduction from $\mathsf{Tot}(\mathsf{U})$ to U, we have that U is Σ_1^0-complete too, as required. □

Since all theories we have introduced are r.e. and modelled by $\mathcal{L}$, we thus immediately have:

Corollary 28. *The equational theories of idempotent Conway μ-semirings, idempotent Park μ-semirings and Chomsky algebras are undecidable, and moreover Σ_1^0-complete.*

5. The case without idempotency

In this section we shall strengthen the main result of the previous section to cover theories of semirings that are not necessarily idempotent.

5.1. **Identity-free μ-semirings and productive expressions.** Write $\mu\mathsf{S}_\chi$ for the axiomatisation defined just like $\mu\mathsf{S}$ but without any of the axioms mentioning 1 (and without 1 in the language).

The **productive** expressions (in the language of $\mu\mathsf{S}_\chi$), written p, q etc., are generated by,

$$p, q, \ldots \quad ::= \quad 0 \quad | \quad a \quad | \quad p + q \quad | \quad pe \quad | \quad ep \quad | \quad \mu X p \tag{5}$$

where e ranges over (possibly open, not necessarily productive) 1-free expressions.

By inspection of the proof of Theorem 7, we can give a more refined result than Theorem 8:

Proposition 29. *A productive CFG has productive solutions in $\mu\mathsf{S}_\chi$. I.e., given a productive CFG $\{X_i \Rightarrow f_{ij}(\vec{X})\}_{i<n,j}$, there are productive expressions $\vec{e} = e_0, \ldots, e_{n-1}$ s.t. $\mu\mathsf{S}_\chi \vdash e_i = \sum_j f_{ij}(\vec{e})$ for all $i < n$.*

In particular note that the RHS of a production in a productive CFG is a productive expression, whence the same proof of Theorem 7 applies for the statement above. Note also that that proof does not rely on any properties of the multiplicative identity 1 (which in any case plays no role in productive CFGs).

5.2. **An idempotent interpretation of productive expressions.** Temporarily write $e' := \mu x(e + \mu y(x + y))$. It is not hard to see that, over $\mu\mathsf{S}_\chi$, e' is an idempotent:

$$\begin{array}{ll} e' = e + \mu y(e' + y) & \text{by unfolding } e', \text{ by axiom (ii)} \\ = e + e' + \mu y(e' + y) & \text{by unfolding } \mu y(e' + y), \text{ by axiom (ii)} \\ = (e + \mu y(e' + y)) + e' & \text{by associativity and commutativity of } + \\ = e' + e' & \text{by the first line} \end{array}$$

In the presence of an order e' is also an upper bound of e, and in the presence of induction e' is in fact the *least* idempotent upper bound of e. In particular in $\mathcal{L}$ (and any other idempotent model) we have simply $e' = e$:

- On one hand, we have $\mathcal{L}(e') = \mathcal{L}(e + \mu y(e' + y))$, and so $\mathcal{L}(e') \supseteq \mathcal{L}(e)$.
- On the other hand, we have that e solves the matrix of e':
  - Clearly $\mathcal{L}(\mu y(e + y)) = e$: as above we have $\mathcal{L}(\mu y(e + y)) = \mathcal{L}(e + \mu y(e + y)) \supseteq \mathcal{L}(e)$; we also have that e solves the matrix of $\mu y(e + y)$ in $\mathcal{L}$, i.e., $\mathcal{L}(e) = \mathcal{L}(e + e)$, thanks to idempotency.
  - Thus we have $e = e + e = e + \mu y(e + y)$ in $\mathcal{L}$, again by idempotency.

  Thus indeed $\mathcal{L}(e') \subseteq \mathcal{L}(e)$, as e' must be a least solution.

Definition 30 (Idempotent interpretation)**.** For 1-free expressions e, define e^i as follows:

- $a^i := a' = \mu x(a + \mu y(x + y))$
- $\cdot^i$ commutes with everything else. I.e.,

$$\begin{array}{rl} 0^i & := 0 \\ x^i & := x \\ (e + f)^i & := e^i + f^i \\ (ef)^i & := e^i f^i \\ (\mu x e)^i & := \mu x e^i \end{array}$$

As previously argued it is not hard to see that:

Proposition 31. $\mathcal{L}(e^i) = \mathcal{L}(e)$.

Proof sketch. By induction on the structure of e. In fact, we must strengthen the statement to $\mathcal{L}(e^i(\vec{A})) = \mathcal{L}(e(\vec{A}))$ for all languages $\vec{A}$. The case when e is some $a \in \mathcal{A}$ follows from the argument above. Every other case is immediate, with the μ-case following by the strong μ-congruence principle in $\mathcal{L}$: $\forall x\, e(x) = f(x) \to \mu x e(x) = \mu x f(x)$. □

Moreover, $\cdot^i$ induces a bona fide interpretation of identity-free μ-semirings:

Proposition 32. $\mu\mathsf{S}_\chi \vdash e = f \implies \mu\mathsf{S}_\chi \vdash e^i = f^i$.

To see this, simply replace every expression e in a $\mu\mathsf{S}_\chi$ proof by e^i. It remains to verify that the axioms and rules remain correct, which is routine.

Finally, thanks to the fact that idempotent elements of a semiring form an ideal, cf. Theorem 12, we have idempotency of the image of $\cdot^i$ on productive expressions provably in $\mu\mathsf{S}_\chi$:

Lemma 33. *If $e(\vec{x})$ is productive then $\mu\mathsf{S}_\chi \vdash e^i(\vec{f}) + e^i(\vec{f}) = e^i(\vec{f})$ for all $\vec{f}$.*

Proof. By induction on the structure of $e(\vec{x})$, according to the grammar of productive expressions in (5). Working in $\mu\mathsf{S}_\chi$ we have:

- If $e(\vec{x})$ is 0 then:

$$\begin{array}{ll} 0^i + 0^i = 0 + 0^i & \text{by definition of } 0^i \\ = 0^i & \text{since 0 is an additive unit} \end{array}$$

- If $e(\vec{x})$ is some $a \in \mathcal{A}$ then $a^i + a^i = a^i$ by the argument at the beginning of this subsection.

- If $e(\vec{x})$ is $p(\vec{x}) + q(\vec{x})$ then:

$$\begin{array}{lll} & (p^i(\vec{f}) + q^i(\vec{f})) + (p^i(\vec{f}) + q^i(\vec{f})) & \\ = & (p^i(\vec{f}) + p^i(\vec{f})) + (q^i(\vec{f}) + q^i(\vec{f})) & \text{by associativity and commutativity of } + \\ = & p^i(\vec{f}) + q^i(\vec{f}) & \text{by inductive hypothesis for } p(\vec{x}) \text{ and } q(\vec{x}) \end{array}$$

- If $e(\vec{x})$ is $p(\vec{x})g(\vec{x})$ then:

$$\begin{array}{lll} & p^i(\vec{f})g^i(\vec{f}) + p^i(\vec{f})g^i(\vec{f}) & \\ = & (p^i(\vec{f}) + p^i(\vec{f}))g^i(\vec{f}) & \text{by distributivity} \\ = & p^i(\vec{f})g^i(\vec{f}) & \text{by inductive hypothesis for } p(\vec{x}) \end{array}$$

- If $e(\vec{x})$ is $g(\vec{x})p(\vec{x})$ then the argument is symmetric to above.
- If $e(\vec{x})$ is $\mu x p(x, \vec{x})$ then:

$$\begin{array}{lll} & \mu x p^i(x, \vec{f}) + \mu x p^i(x, \vec{f}) & \\ = & p^i(\mu x p^i(x, \vec{f}), \vec{f}) + p^i(\mu x p^i(x, \vec{f}), \vec{f}) & \text{by } \mu\text{-unfolding, axiom (ii)} \\ = & p^i(\mu x p^i(x, \vec{f}), \vec{f}) & \text{by inductive hypothesis for } p(x, \vec{x}) \quad \square \end{array}$$

Note that, the above result notwithstanding, e^i is typically not productive, even if e is. In particular a^i is not productive, even though a is.

Notice that, Theorem 32 notwithstanding, $\cdot^i$ does not necessarily interpret $\mu\mathsf{S}_\gamma + \mathsf{idem}$ into $\mu\mathsf{S}_\gamma$. This is because, a priori, a proof in $\mu\mathsf{S}_\gamma + \mathsf{idem}$ may exploit idempotency of expressions not in the image of $\cdot^i$ on productive expressions. For this reason, in what follows, we shall exploit the fact that we carefully kept track of the use of idempotency in our arguments in the previous section.

5.3. **Undecidability of $\mu\mathsf{S}_\gamma$ (and extensions).** Recall $\top := \mu x \left(\sum \mathcal{A} + \sum_{a \in \mathcal{A}} ax \right)$. By a similar argument to Theorem 27 in the previous section we can obtain:

Proposition 34. *Let $G \in \mathbf{R}$ and let e_X be productive solutions of each of its non-terminals X in $\mu\mathsf{S}_\gamma$, cf. Theorem 29, with $e_T = \top$. Then $\mu\mathsf{S}_\gamma \vdash \top^i \leq e_S^i$.*

For this we need appropriate versions of the two intermediate results we employed, Theorems 25 and 26, in the previous section. Their proofs are morally similar to the previous ones, but we must be careful to state them appropriately in the absence of idempotency, exploiting the refined results of this section on productive CFGs and $\mu\mathsf{S}_\gamma$

Lemma 35. *Fix a productive CFG G and write e_X for (any) corresponding productive solutions in $\mu\mathsf{S}_\gamma$ of each non-terminal X of G, cf. Theorem 29. If $\vec{a} \in \mathcal{L}(X)$ in G then $\mu\mathsf{S}_\gamma \vdash \vec{a}^i \leq e_X^i$.*

Proof. By induction on the derivation tree of $\vec{a}$ from X in G. Working in $\mu\mathsf{S}_\gamma$:

- If $X \to \vec{a}$ then we have $\vec{a} + g = e_X$ for some g by Theorem 29. Thus $\vec{a}^i + g^i = e_X^i$ by Theorem 32, and so $\vec{a}^i \leq e_X^i$ by idempotency of $\vec{a}^i$, Theorem 33.
- The inductive step is the same as in the proof of Theorem 25. $\square$

Lemma 36. *For $n > 0$, we have $\mu\mathsf{S}_\gamma \vdash \top^i = \sum_{|\vec{a}|=1}^{n} \vec{a}^i + \sum_{|\vec{a}|=n} \vec{a}^i \top^i$.*

Proof. The proof of Theorem 26 does not use any identity axioms, and so goes through already in $\mu\mathsf{S}_\gamma$. The result now follows by Theorem 32. $\square$

We are now ready to prove the result we stated at the beginning of this section:

Proof of Theorem 34. The proof goes through just like the proof of Theorem 27, only replacing all (productive) expressions by their image under $\cdot^i$. All appeals to Theorems 25 and 26 are replaced by Theorems 35 and 36 respectively. Note that, by inspection of the proof, the only uses of idempotency (marked in purple) are on productive expressions, whose image under $\cdot^i$ is indeed provably idempotent in $\mu\mathsf{S}_\gamma$ by Theorem 33. In particular, the expressions e_Y and e_S are productive by assumption. $\square$

5.4. **Putting it together.** Finally we arrive at our promised strengthening of Theorem 24:

Theorem 37. *Any r.e. extension of $\mu\mathsf{S}_\gamma$ sound for $\mathcal{L}$ is Σ_1^0-complete.*

The proof is similar to that of Theorem 24, only being more careful about handling the $\cdot^i$ translation.

Proof. Like in the proof of Theorem 24, let us assume that all CFGs have start symbol S and write e_X^G for the productive solutions of nonterminals X in a productive grammar G obtained by Theorem 29. Let $\mathsf{U} \supseteq \mu\mathsf{S}_\gamma$ be r.e. with $\mathcal{L} \models \mathsf{U}$. Now we write $\mathsf{Tot}(\mathsf{U}) := \{G \text{ productive} : \mathsf{U} \vdash \top^i \leq (e_S^G)^i\}$. We have:

- $\mathsf{Tot}(\mathsf{U}) \cap \bar{\mathbf{T}} = \varnothing$, by assumption that $\mathcal{L} \models \mathsf{U}$ and by Theorem 31.

- $\mathbf{R} \subseteq \mathsf{Tot}(\mathsf{U})$, since $\mu\mathsf{S}_\chi \subseteq \mathsf{U}$ and by Theorem 34.

Thus by Theorems 18 and 20 we have that $\mathsf{Tot}(\mathsf{U})$ is Σ^0_1-complete. Again, by the obvious computable reduction from $\mathsf{Tot}(\mathsf{U})$ to U, we conclude as required. □

Corollary 38. *The equational theories of Conway μ-semirings and Park μ-semirings are undecidable, and moreover Σ^0_1-complete. The same holds even for identity-free versions of these theories.*

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